National University of Computer and Emerging Sciences

*Project Phase – Iteration 1*

# CryptoWise – AI Crypto Assistant

*Software Engineering*

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# Chapter 1: Project Introduction

## 1.1 Project Title

CryptoWise – AI Powered Crypto Prediction & Trading Platform

## 1.2 Project Description

CryptoWise proposes an intelligent cryptocurrency investment assistant that leverages historical crypto market data obtained from datasets to train a machine learning model. The model analyzes user-defined parameters and market indicators to generate predictions and insights related to cryptocurrency investments. To enhance usability, a chatbot is integrated, enabling users to interact with the system through natural language conversations. This combination of predictive analytics and conversational AI aims to make crypto investment analysis more accessible, personalized, and informative for users.

## 1.3 Sprint 1 – Module Focus

Sprint 1 focuses on the Core Platform module - the foundational layer of the CryptoWise application. This sprint establishes all essential infrastructure including user authentication, the main dashboard homepage, live cryptocurrency price display, the prices page with multi asset support, and a responsive UI. These components form the backbone upon which all future ML prediction and trading bot features will be built.

## 1.4 Tech Stack

- Frontend: **Next.js** (React-based full-stack framework)
- Backend: **Python** (Fast API / Flask for ML and API services)
- Database: **MongoDB** for user and market data
- Real-Time Data: **Binance** WebSocket API, Gold API

# Chapter 2: Sprint Backlog – Sprint 1

## 2.1 Module Name

*Module:* Core Platform – Authentication, Homepage, Live Prices & Responsive UI

## 2.2 Sprint Duration

*Sprint 1 Duration:* 2 Weeks

## 2.3 Sprint Backlog Table

| Story ID | User Story Title | Priority | Est. Hours | Status |
|---|---|---|---|---|
| US-01 | User Authentication | High | 10 | Completed |
| US-02 | Homepage & Hero Section | High | 8 | Completed |
| US-03 | Live Price Ticker | High | 12 | Completed |
| US-04 | Prices Page with Search | High | 10 | Completed |
| US-05 | Responsive UI | Medium | 8 | Completed |
| US-06 | About Page with Team Info | Low | 4 | Completed |

# Chapter 3: User Stories

The following user stories have been selected for Sprint 1 development.

| **Story ID: US-01** **Story Title: User Authentication** | **Priority: High**<br>**Estimated Hours: 10** |
|---|---|
| **User Story:**<br>As a user, I want to register and log in to my account so that I can access personalized features and secure my data.<br>**Description:**<br>This story covers the complete authentication flow including user registration with email and password, secure login with JWT tokens, password validation, and session management. It ensures only authorized users can access the CryptoWise platform.<br>**Sub User Stories:**<br>• As a new user, I want to register using my email and password.<br>• As a registered user, I want to log in securely to access my dashboard.<br>• As a user, I want to log out and have my session terminated.<br>• As a user, I want to receive an error message if my credentials are incorrect.<br>**Acceptance Criteria:**<br>• Registration form must validate email format and password strength (min 8 characters).<br>• Successful login must redirect user to the main dashboard.<br>• JWT token must be stored securely and expire after 24 hours.<br>• Incorrect credentials must display a clear error message without revealing which field is wrong.<br>• Logout must clear all session data and redirect to the login page. | |

| **Story ID: US-02** **Story Title: Homepage & Hero Section** | **Priority: High**<br>**Estimated Hours: 8** |
|---|---|
| **User Story:**<br>As a visitor, I want to view an informative and visually engaging homepage so that I can understand what CryptoWise offers and get started quickly.<br>**Description:**<br>This story involves designing and developing the main landing page of CryptoWise. It includes a hero section with key statistics, a | |

navigation bar, feature highlights, and call-to-action buttons that guide users to register or explore the platform.

**Sub User Stories:**

- As a visitor, I want to see a hero section with platform statistics (accuracy rate, active bots, users).
- As a visitor, I want to see a navigation bar with links to all major sections.
- As a visitor, I want to click a 'Get Started' button that takes me to the registration page.
- As a visitor, I want to see a summary of platform features on the homepage.

**Acceptance Criteria:**

- Hero section must display accuracy rate (94%), active bots count, and user count.
- Navigation bar must include links to Home, Prices, Predictions, Bots, Assistant, and About.
- 'Get Started' and 'Live Demo' buttons must be visible and functional.
- Homepage must load within 2 seconds on a standard internet connection.
- All navigation links must be accessible and correctly routed.

| **Story ID: US-03** **Story Title: Live Price Ticker** | **Priority: High**<br>**Estimated Hours: 12** |
|---|---|
| **User Story:**<br>As a user, I want to view live cryptocurrency and precious metal prices with real-time updates so that I can monitor the market at a glance.<br>**Description:**<br>This story covers the implementation of a real-time price ticker that streams live market data from Binance WebSocket API (for crypto) and Gold API (for metals). Prices update every 5 seconds and show 24-hour high, low, and percentage change data.<br>**Sub User Stories:**<br>• As a user, I want to see live prices for BTC, ETH, BNB, SOL, XRP, Gold, and Silver.<br>• As a user, I want prices to update automatically every 5 seconds without refreshing the page.<br>• As a user, I want to see the 24-hour price change percentage with color indicators (green/red).<br>• As a user, I want to see the 24-hour high and low values for each asset. | |

| | |
|---|---|
| **Acceptance Criteria:**<br>• Price ticker must display real-time data for all 7 assets (5 crypto + 2 metals).<br>• WebSocket connection must update prices every 5 seconds automatically.<br>• Positive price changes must be displayed in green; negative changes in red.<br>• Ticker must reconnect automatically if the WebSocket connection is lost.<br>• Price data must include: current price, 24h change %, 24h high, 24h low. | |

| **Story ID: US-04 Story Title: Prices Page with Search** | **Priority: High**<br>**Estimated Hours: 10** |
|---|---|
| **User Story:**<br>As a user, I want to view a dedicated market prices page with search functionality so that I can quickly find and monitor specific assets.<br>**Description:**<br>This story involves building a comprehensive market prices page that displays all tracked assets in a structured table format. Users can search for specific assets by name or symbol. The page also shows trading volume data and is updated in real time.<br>**Sub User Stories:**<br>• As a user, I want to see all assets listed in a sortable table with prices and statistics.<br>• As a user, I want to search for a specific asset by name or symbol.<br>• As a user, I want to see trading volume data for each cryptocurrency.<br>• As a user, I want the table to update in real time without page refresh.<br>**Acceptance Criteria:**<br>• Prices page must display: asset name, current price, 24h change %, 24h high, 24h low, and volume.<br>• Search bar must filter assets in real time as the user types.<br>• Table must display data for all 7 tracked assets.<br>• Volume data must be displayed for cryptocurrencies (N/A for metals).<br>• Page must be responsive and readable on both desktop and mobile screens. | |

| **Story ID: US-05** **Story Title: Responsive UI** | **Priority: Medium**<br>**Estimated Hours: 8** |
|---|---|
| **User Story:**<br>As a user, I want the platform to have a responsive dark mode interface so that I can use it comfortably across devices in any lighting environment.<br>**Description:**<br>This story covers the implementation of a fully responsive user interface with dark mode as the default theme. The UI must adapt seamlessly to different screen sizes (desktop, tablet, mobile) and maintain consistent styling across all pages using Tailwind CSS.<br>**Sub User Stories:**<br>• As a user, I want the platform to default to dark mode for a professional trading experience.<br>• As a user, I want the interface to be fully responsive on mobile and tablet devices.<br>• As a user, I want consistent styling and color themes across all pages.<br>• As a user, I want the navigation to collapse into a hamburger menu on mobile screens.<br>**Acceptance Criteria:**<br>• Dark mode must be applied consistently across all pages and components.<br>• UI must be fully responsive and functional on - screen widths from 320px to 2560px.<br>• Navigation must display a hamburger menu on screens narrower than 768px.<br>• Color contrast ratios must meet WCAG 2.1 AA accessibility standards.<br>• Page layout must not break or overflow on any standard screen size. | |

| **Story ID: US-06** **Story Title: About Page with Team Info** | **Priority: Low**<br>**Estimated Hours: 4** |
|---|---|
| **User Story:**<br>As a visitor, I want to view an About page with information about the CryptoWise team so that I can understand the people behind the platform.<br>**Description:**<br>This story involves creating a simple yet professional About page that showcases the project team members, their roles, and brief descriptions of the technologies used. This adds credibility to the platform and provides transparency to users. | |

| | |
|---|---|
| **Sub User Stories:**<br>• As a visitor, I want to see team member names, roles, and photos on the About page.<br>• As a visitor, I want to see a brief description of the CryptoWise project and its mission.<br>• As a visitor, I want to see the technology stack used to build the platform.<br>**Acceptance Criteria:**<br>• About page must list all 3 team members with their names, roll numbers, and assigned roles.<br>• A project mission statement must be displayed on the About page.<br>• Technology stack section must list all major technologies used (Next.js, Python, etc.).<br>• Page must follow the same dark mode styling as the rest of the platform.<br>• About page must be accessible from the main navigation bar. | |

# Chapter 4: Structured Specifications

The following structured specifications detail the functional, UX, security, and performance requirements for each sub user story. These specifications serve as the precise implementation contract for all Sprint 1 features.

## 4.1 Structured Specifications for US-01: User Authentication

***US-01.1:*** *User Registration with Email and Password*

| Type | Specification |
|---|---|
| **Functional** | The system must provide a registration form with fields for full name, email address, and password. |
| **Functional** | The system must validate that the email is in correct format before submission. |
| **Functional** | Password must be at least 8 characters long and contain at least one uppercase letter and one number. |
| **Security** | User passwords must be hashed using encrypt before being stored in the database. |
| **UX** | Inline validation messages must appear immediately below the relevant field when an error is detected. |

***US-01.2:*** *Secure Login*

| Type | Specification |
|---|---|
| **Functional** | The system must authenticate users using email and password and return a signed JWT token upon success. |
| **Security** | JWT tokens must expire after 24 hours and be stored securely in HTTP-only cookies. |
| **Functional** | If credentials are invalid, the system must return a generic error message without specifying which field is incorrect. |
| **Performance** | Login authentication process must complete within 1 second under normal server load. |

***US-01.3:*** *Logout and Session Termination*

| Type | Specification |
|---|---|
| **Functional** | Logging out must invalidate the JWT token and clear all session cookies on the client side. |
| **Security** | After logout, accessing any protected route must redirect the user to the login page. |
| **UX** | A logout confirmation must not be required; logout must happen instantly on button click. |

### *US-01.4: Error Handling for Incorrect Credentials*

| Type | Specification |
|---|---|
| **Functional** | The system must display an error message when login fails due to wrong email or password. |
| **Security** | Error messages must not reveal whether the email or the password was incorrect. |
| **UX** | Error messages must be displayed in red text directly below the login form fields. |

## 4.2 Structured Specifications for US-02: Homepage & Hero Section

### *US-02.1: Hero Section with Platform Statistics*

| Type | Specification |
|---|---|
| **Functional** | The hero section must display platform statistics including ML accuracy rate (94%), number of active bots (100+), and total users. |
| **UX** | Statistics must be displayed as animated counters that increment to their final values on page load. |
| **Performance** | The homepage must fully render within 2 seconds on a standard broadband connection. |

### *US-02.2: Navigation Bar with Links*

| Type | Specification |
|---|---|
| **Functional** | The navigation bar must include working links to: Home, Prices, Predictions, Predictor, Bots, Assistant, and About pages. |
| **UX** | The currently active page link must be visually highlighted in the navigation bar. |
| **Functional** | A Login button must be visible in the navigation bar for unauthenticated users, and a Logout option for authenticated users. |

### *US-02.3: Call-to-Action Buttons*

| Type | Specification |
|---|---|
| **Functional** | The 'Get Started' button must redirect unauthenticated users to the registration page. |
| **Functional** | The 'Live Demo' button must navigate users to the Prices page to explore live market data. |
| **UX** | Buttons must have hover animations and be clearly distinguishable from surrounding content. |

## 4.3 Structured Specifications for US-03: Live Price Ticker

### *US-03.1: Real-Time Price Display for All Assets*

| Type | Specification |
|---|---|
| **Functional** | The ticker must display live prices for BTC, ETH, BNB, SOL, XRP and Gold, Silver. |
| **Functional** | Each asset entry must show: asset name/icon, current price in USD, 24h change percentage, 24h high, and 24h low. |
| **Performance** | Price data must be streamed via WebSocket and updated on the UI every 5 seconds without requiring a page refresh. |

### *US-03.2: Color-Coded Price Change Indicators*

| Type | Specification |
|---|---|
| **UX** | Positive 24h price changes must be displayed in green and negative changes in red. |
| **UX** | An upward or downward arrow icon must accompany the percentage change value to reinforce the direction. |
| **Functional** | Percentage change values must be calculated accurately relative to the 24h opening price. |

### *US-03.3: WebSocket Connection Management*

| Type | Specification |
|---|---|
| **Functional** | The system must establish a WebSocket connection to Binance on page load and maintain it throughout the session. |
| **Functional** | If the WebSocket connection drops, the system must attempt to reconnect automatically with exponential backoff. |
| **UX** | A connection status indicator must be visible to inform users whether live data is active or reconnecting. |

## 4.4 Structured Specifications for US-04: Prices Page with Search

### *US-04.1: Asset Table with Full Market Data*

| Type | Specification |
|---|---|
| **Functional** | The prices table must display all 7 assets with columns: Asset, Price, 24H Change, 24H High, 24H Low, Volume. |
| **Functional** | Volume data must be shown for cryptocurrencies and displayed as 'N/A' for precious metals. |
| **UX** | Table rows must have alternating background colors for readability, with hover highlighting. |

***US-04.2:*** *Real-Time Search Functionality*

| Type | Specification |
|---|---|
| **Functional** | A search bar must be available at the top of the prices page that filters visible assets in real time as the user types. |
| **Functional** | Search must match against both the asset full name and the symbol. |
| **UX** | If no assets match the search query, a 'No results found' message must be displayed in place of the table. |

## 4.5 Structured Specifications for US-05: Responsive UI

***US-05.1:*** *Default Dark Mode Theme Application*

| Type | Specification |
|---|---|
| **Functional** | Dark mode must be applied as the default theme across all pages and components on initial load. |
| **UX** | Background colors must use dark shades and text must use high contrast light colors. |
| **Performance** | Theme must be applied before first render to prevent any flash of un style content. |

***US-05.2:*** *Responsive Layout Across Devices*

| Type | Specification |
|---|---|
| **Functional** | All pages must be fully functional and visually complete on - screen widths from 320px to 2560px. |
| **UX** | Navigation must collapse into a hamburger menu icon on screens narrower than 768px. |
| **UX** | Data tables must become horizontally scrollable on mobile screens to avoid content overflow. |

## 4.6 Structured Specifications for US-06: About Page

***US-06.1:*** *Team Members Display*

| Type | Specification |
|---|---|
| **Functional** | The About page must list all 3 team members with their full names, roll numbers, and assigned project roles. |
| **UX** | Each team member entry must be displayed in a styled card component consistent with the platform's dark theme. |
| **Functional** | Project supervisor name must also be displayed on the About page. |

### ***US-06.2:*** *Project Mission and Technology Stack*

| Type | Specification |
|---|---|
| **Functional** | A concise project mission statement describing CryptoWise's purpose and goals must be displayed. |
| **Functional** | The technology stack section must list all major technologies: Next.js, Python, Binance API, Gold API, GROQ, LSTM, XG Boost. |
| **UX** | Technology stack items must be displayed as styled badge components with appropriate icons or logos. |

# Chapter 5: Scrum Board – Trello Snapshots

The team utilized Trello (https://trello.com/) to manage and track the Sprint 1 backlog. Three snapshots were captured at different stages of the sprint to document progress. The Trello board was organized into four columns: To Do, In Progress, In Review, and Done.

## 5.1 Snapshot 1 – Sprint Start (Backlog Uploaded)

This snapshot was taken at the beginning of Sprint 1 when all user stories were uploaded to the Trello board. All 6 user stories were placed in the 'To Do' column. The team had assigned story points, priorities, and responsible members to each card.

## 5.2 Snapshot 2 – Mid Sprint (Work in Progress)

This snapshot was taken at the midpoint of Sprint 1. US-01 (Authentication) and US-02 (Homepage) had been moved to 'Done'. US-03 (Live Price Ticker) was 'In Progress'. US-04, US-05, and US-06 remained in 'To Do'. Approximately 50% of planned work had been completed at this stage.

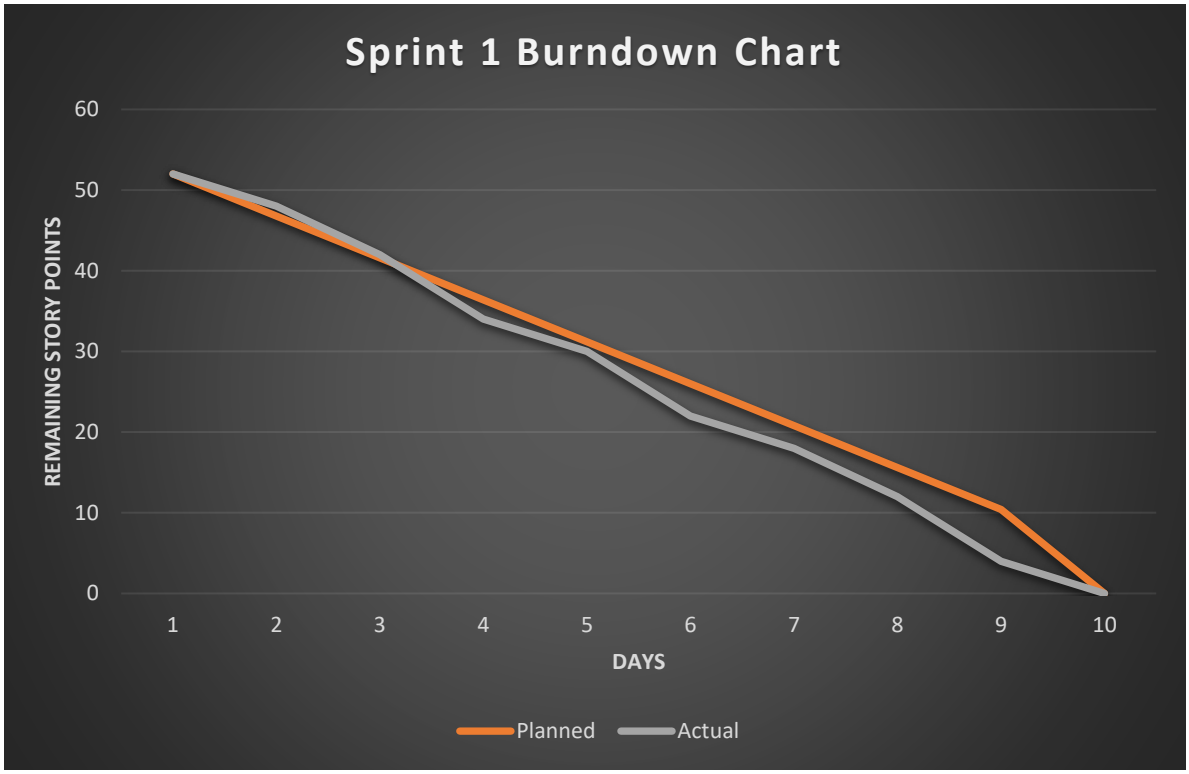
## 5.3 Snapshot 3 – Sprint End (Completed / Leftovers)

This snapshot was taken at the conclusion of Sprint 1. All 6 user stories were successfully completed and moved to the 'Done' column. There were no leftover stories for Sprint 2. The team completed all planned deliverables within the 2 - week sprint timeline.

# Chapter 6: Product Burn - Down Chart – Sprint 1

The burn - down chart below tracks planned story points versus completed story points over the 10-day Sprint 1 period. The ideal burn-down line shows the expected linear completion rate, while the actual line reflects real progress. Total sprint points: 52.

| Day | Planned Remaining | Actual Remaining | Tasks Planned | Tasks Completed |
|---|---|---|---|---|
| Day 1 | 52 | 52 | 0 | 0 |
| Day 2 | 46.8 | 48 | US-01 start | Auth setup |
| Day 3 | 41.6 | 42 | US-01 end | US-01 done |
| Day 4 | 36.4 | 34 | US-02 start | US-02 done |
| Day 5 | 31.2 | 30 | US-03 start | US-03 partial |
| Day 6 | 26 | 22 | US-03 end | US-03 done |
| Day 7 | 20.8 | 18 | US-04 start | US-04 done |
| Day 8 | 15.6 | 12 | US-05 start | US-05 done |
| Day 9 | 10.4 | 4 | US-06 start | US-06 done |
| Day 10 | 0 | 0 | Review & QA | Sprint complete |



# Chapter 7: Non-Functional Requirements (NFRs)

The following non - functional requirements define the quality attributes and operational constraints of the CryptoWise platform. These are specified using natural language and include measurable benchmarks to ensure the system meets professional trading platform standards.

## 7.1 Performance

The CryptoWise platform must deliver fast and responsive data updates to support real-time trading decisions. All live price data must be fetched and rendered on the UI within 500 milliseconds of the WebSocket update event. Page load times for the main dashboard and prices page must not exceed 2 seconds on a standard broadband connection (25 Mbps or higher). The platform must sustain concurrent WebSocket connections for at least 1,000 simultaneous users without degradation in update frequency. API response times from the backend for ML prediction requests must remain under 3 seconds for single-asset predictions and under 5 seconds for multi-asset batch predictions. Any UI interaction such as searching for an asset or switching pages must respond within 100 milliseconds to maintain a seamless user experience.

## 7.2 Usability

The CryptoWise platform must be intuitive enough for users with no prior trading experience to navigate and understand core features without requiring external documentation. A first-time user must be able to register, log in, and view live prices within 3 minutes of arriving at the platform homepage. All interactive elements including buttons, form fields, and navigation links must provide clear visual feedback such as hover states, loading indicators, and error messages. Error messages must be written in plain language, clearly describing what went wrong and how to resolve it. The platform must provide tooltips or labels for technical financial terms such as ATR, RSI, and MACD to assist beginner traders. The interface must maintain a consistent visual language across all pages with standardized typography, color usage, and component spacing.

## 7.3 Security

The CryptoWise platform must implement industry-standard security measures to protect user data and financial interactions. All user passwords must be hashed using the b crypt algorithm with a minimum cost factor of 12 before being stored in the database. Authentication must be handled using JSON Web Tokens (JWT) with an expiration period of 24 hours, stored in HTTP-only cookies to prevent XSS attacks. All API endpoints must be protected and require a valid JWT token; unauthorized requests must receive a 401 response. The platform must enforce HTTPS for all client-server communication to prevent data interception. Input fields must be sanitized server-side to prevent SQL injection and cross-site scripting (XSS) attacks. Rate limiting must be applied to login and registration endpoints, allowing a maximum of 10 requests per minute per IP address to prevent brute-force attacks.

## 7.4 Scalability

The CryptoWise system must be architected to support growth in both user base and data volume without requiring significant redesign. The backend must be capable of handling at least 10,000 concurrent users during peak market hours without service degradation. The database schema must support the addition of new cryptocurrency assets and precious metals without structural changes to existing tables. The ML model serving infrastructure must be designed to allow new models (beyond

LSTM and XG Boost) to be integrated as independent modules without modifying the core prediction pipeline. The platform must support horizontal scaling of backend services using containerization (Docker) to enable deployment on cloud infrastructure. Data storage for historical price records must efficiently handle at least 5 years of hourly OHLCV data for all 7 tracked assets, amounting to approximately 300,000 records, with query times under 500 milliseconds.

## 7.5 Reliability

The CryptoWise platform must operate with high availability to ensure traders have continuous access to market data and tools. The system must achieve an uptime of at least 99.5% during any calendar month, excluding scheduled maintenance windows. In the event of a WebSocket disconnection from the Binance or Gold API, the system must automatically attempt to reconnect within 5 seconds using an exponential backoff strategy, with a maximum of 5 retry attempts. If the ML prediction service becomes unavailable, the platform must continue to function and display live prices while showing an informative error message for the prediction features. All critical user data including account information and portfolio settings must be backed up daily to prevent data loss. The system must implement graceful error handling throughout the application so that a failure in one module (e.g., the bot service) does not cause a crash in unrelated modules (e.g., the prices page).

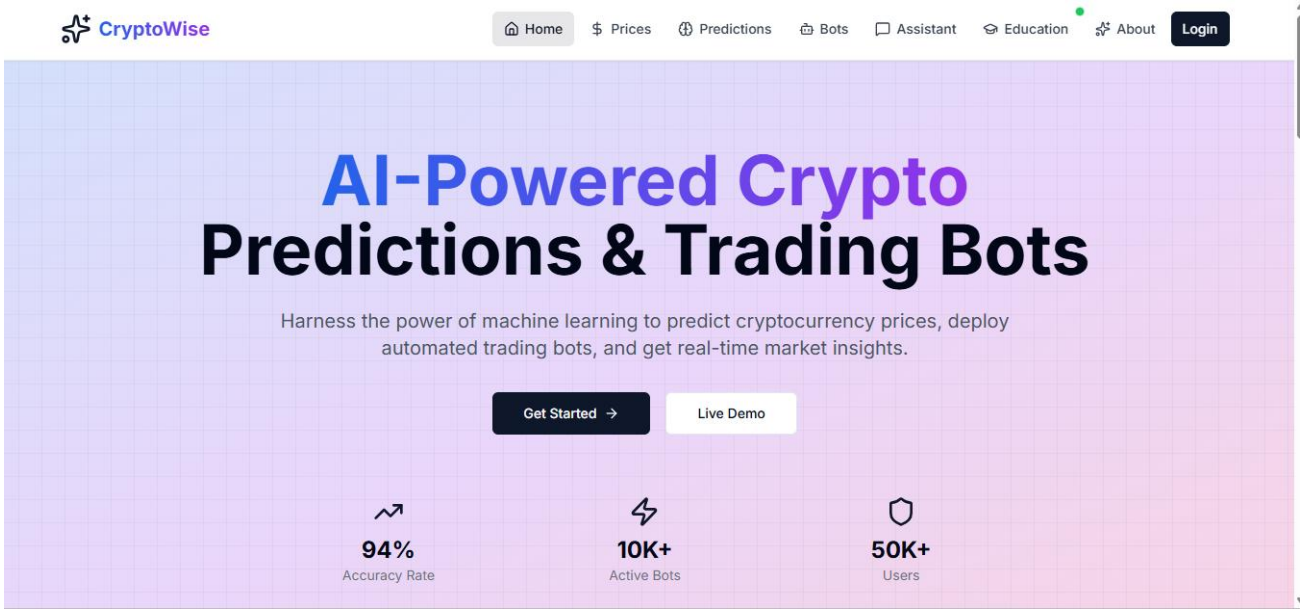
## 7.6 Compatibility

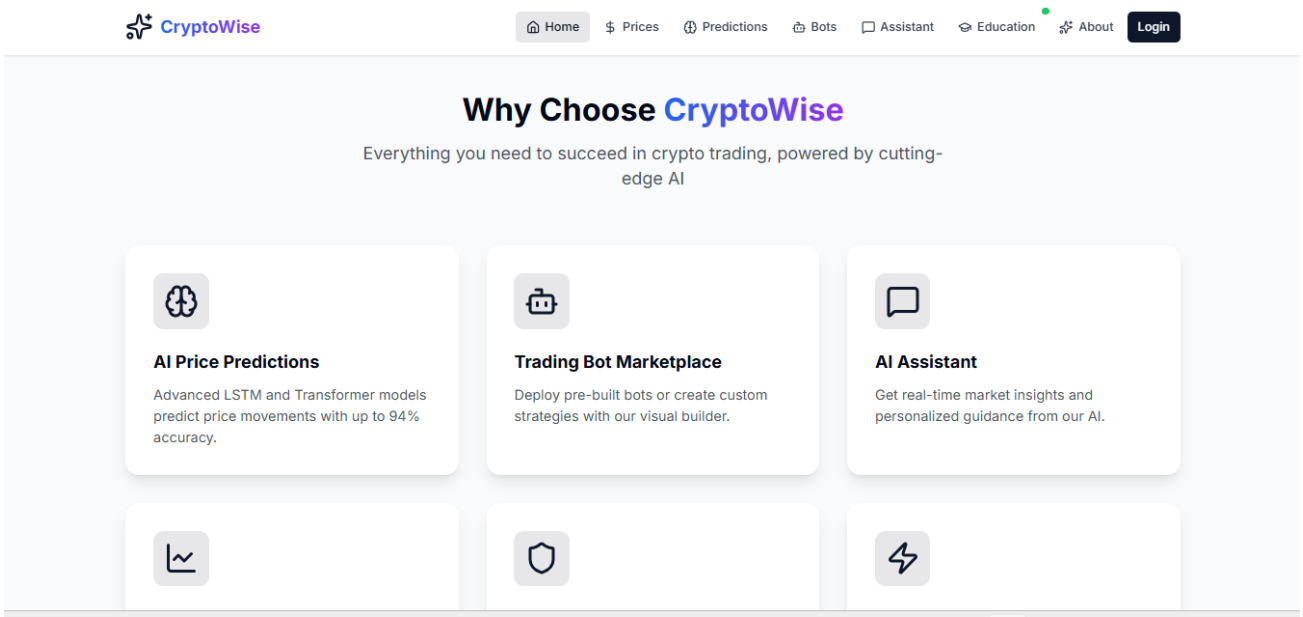
The CryptoWise platform must function correctly and consistently across all modern web browsers and operating systems. The application must fully support the latest versions of Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari, with no loss of functionality or visual inconsistencies. The platform must be fully operational on Windows 10+, macOS 11+, and Ubuntu 20.04+ desktop environments. On mobile devices, the platform must render correctly on iOS 14+ (Safari) and Android 10+ (Chrome) and provide a fully usable experience on screen resolutions as small as 375x667 pixels. The Next.js frontend must maintain compatibility with Node.js version 18 LTS or higher. Third-party integrations including Binance WebSocket API, Gold API, and GROQ AI must be handled with versioned API calls to ensure stability against upstream changes.

# Chapter 8: Iteration 1 – Implementation Screenshots

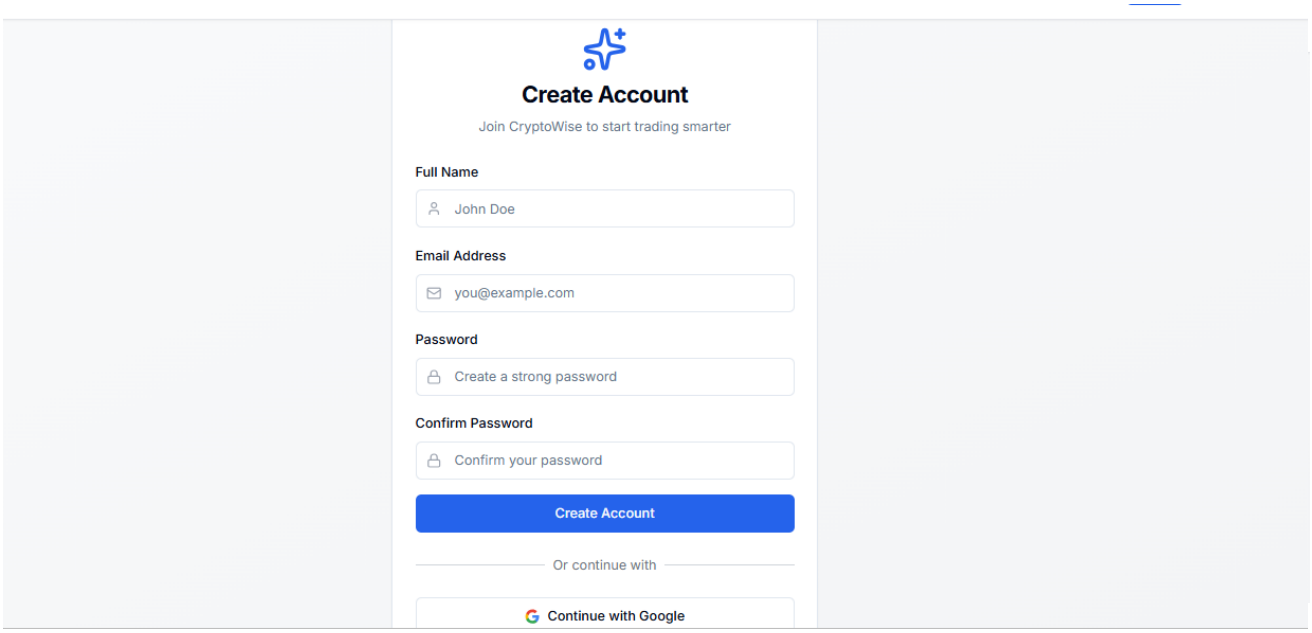
The following screenshots document the actual implemented state of the CryptoWise platform at the end of Sprint 1 / Iteration 1. All 6 user stories have been successfully implemented and the core platform is fully functional.

## 8.1 Homepage Section

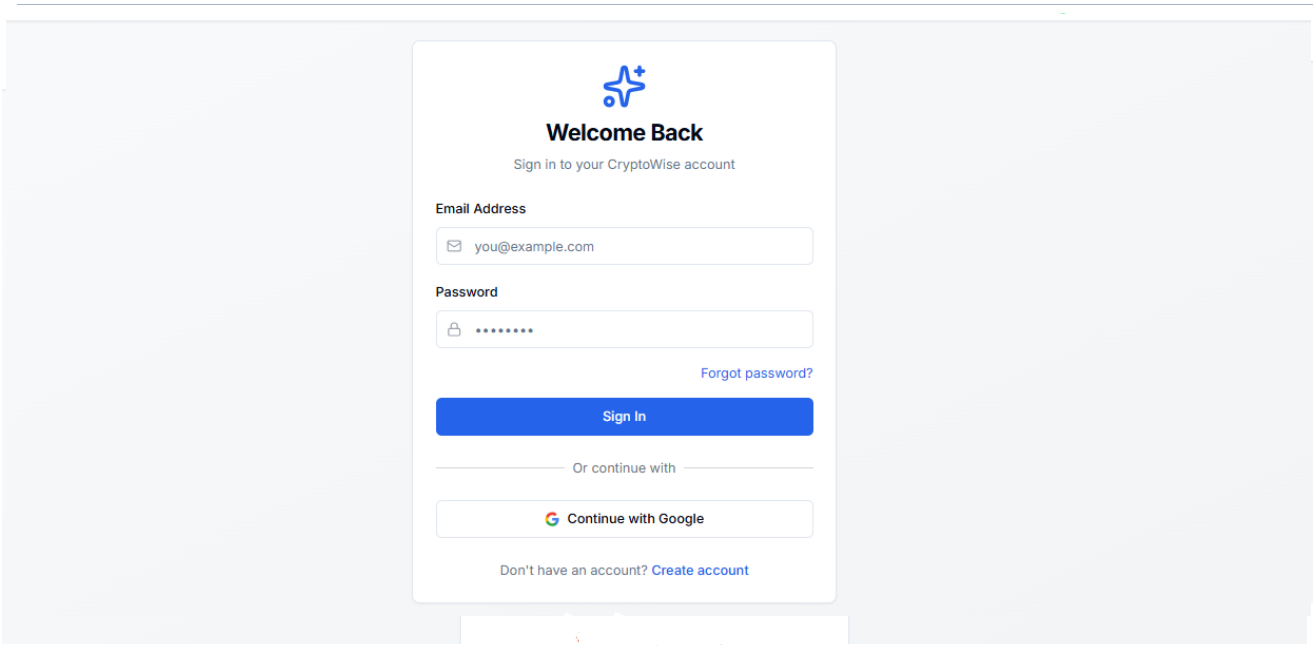


## 8.2 User Registration Page

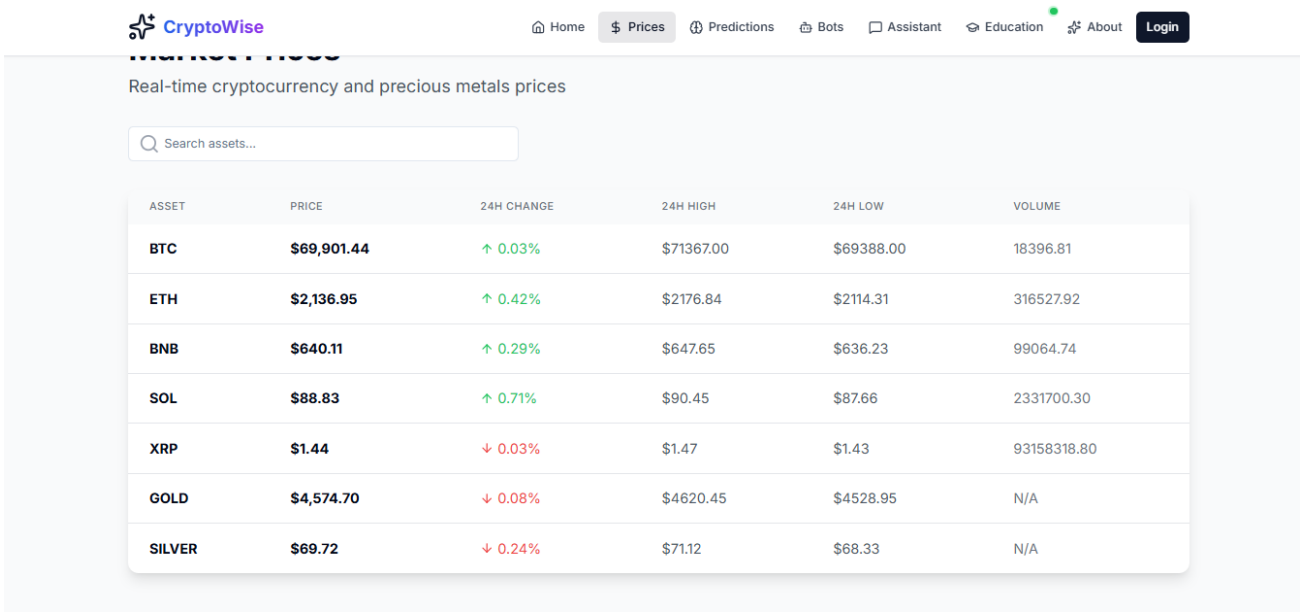



## 8.3 User Login Page



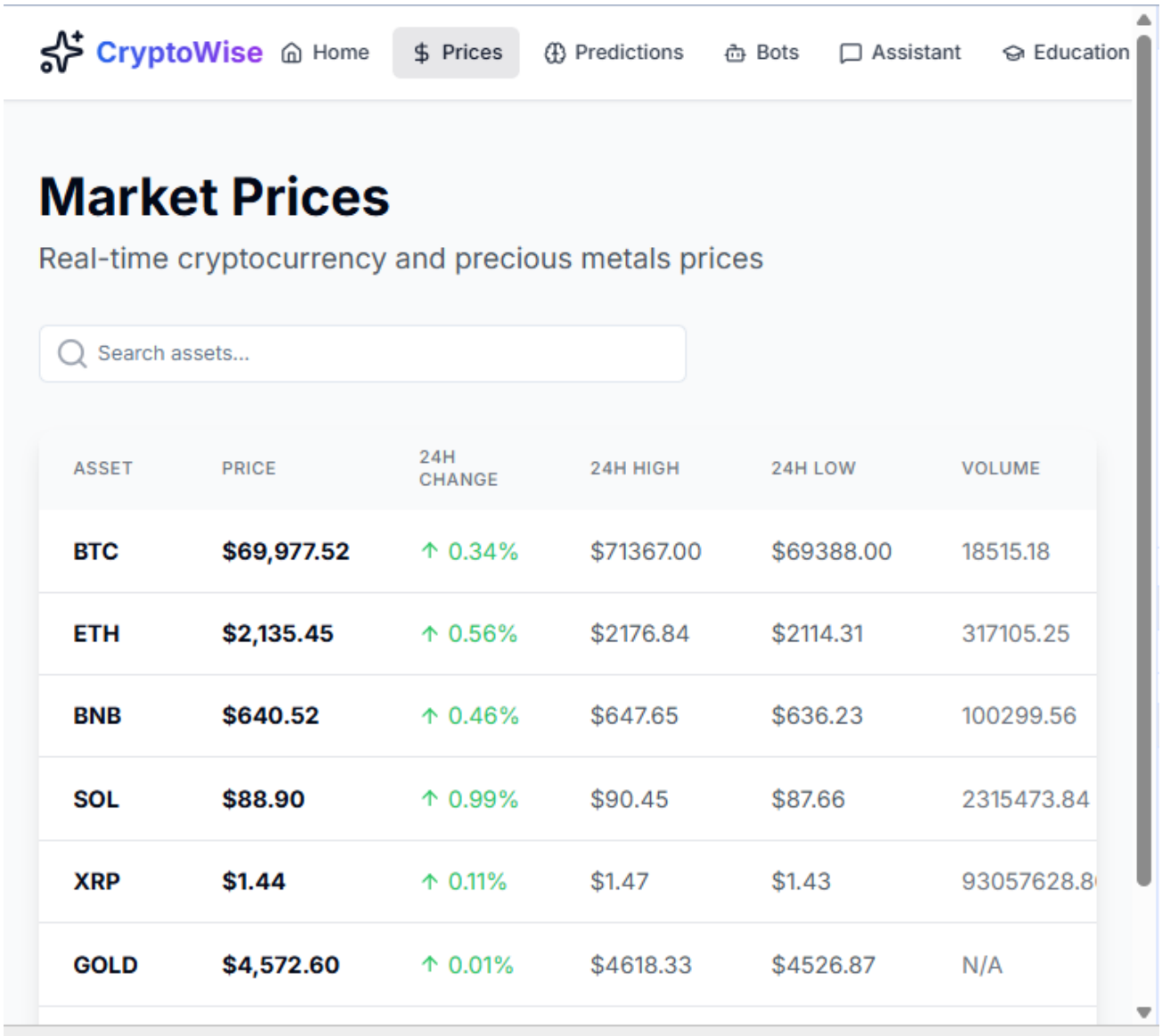
## 8.4 Live Price Ticker

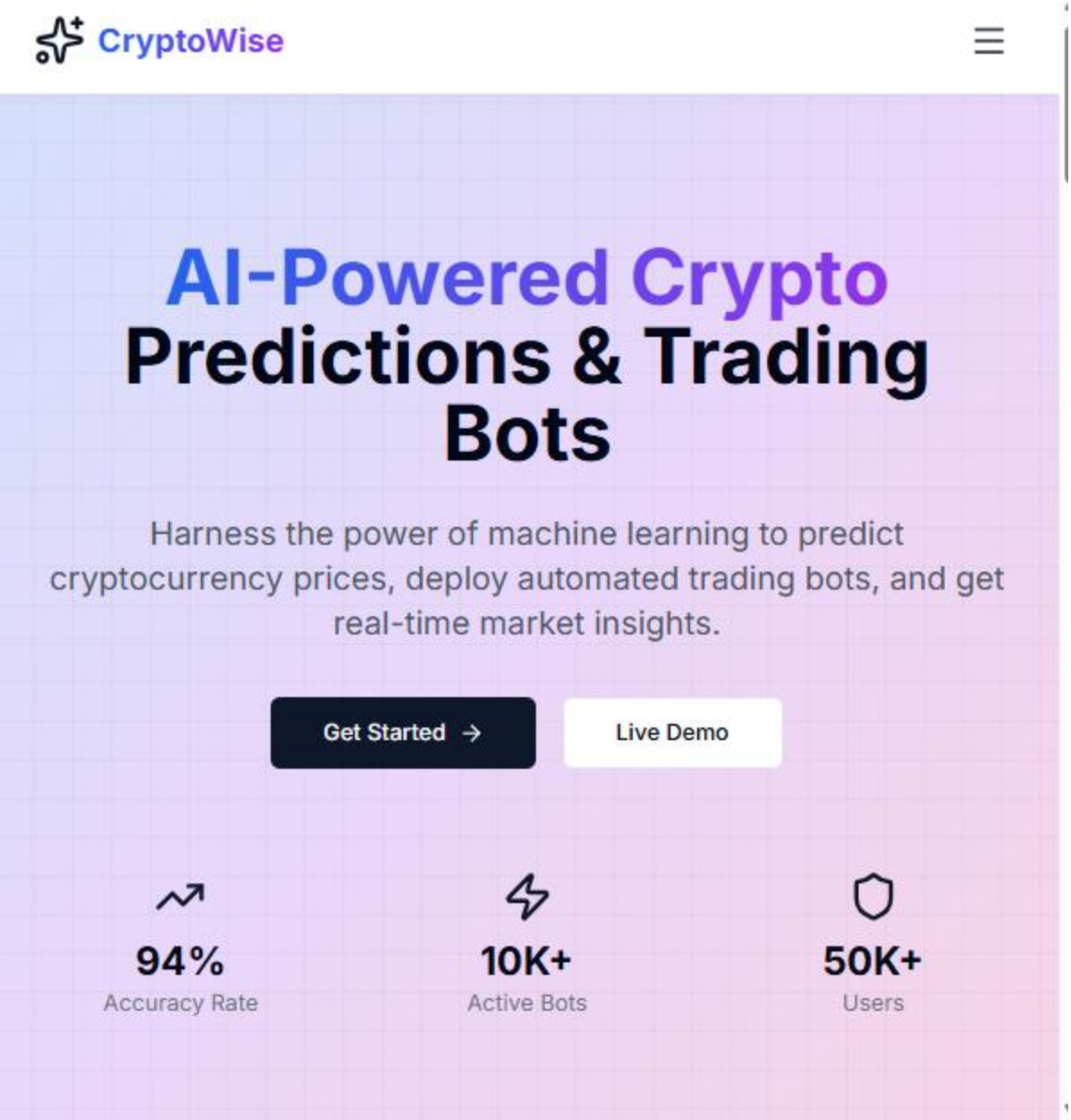
## 8.5 Market Prices Page




| ASSET | PRICE | 24H CHANGE | 24H HIGH | 24H LOW | VOLUME |
|---|---|---|---|---|---|
| BTC | $69,901.44 | ↑ 0.03% | $71367.00 | $69388.00 | 18396.81 |
| ETH | $2,136.95 | ↑ 0.42% | $2176.84 | $2114.31 | 316527.92 |
| BNB | $640.11 | ↑ 0.29% | $647.65 | $636.23 | 99064.74 |
| SOL | $88.83 | ↑ 0.71% | $90.45 | $87.66 | 2331700.30 |
| XRP | $1.44 | ↓ 0.03% | $1.47 | $1.43 | 93158318.80 |
| GOLD | $4,574.70 | ↓ 0.08% | $4620.45 | $4528.95 | N/A |
| SILVER | $69.72 | ↓ 0.24% | $71.12 | $68.33 | N/A |

### 8.6 Responsive Mobile View

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# Chapter 9: Work Division

The following table outlines the specific tasks completed by each team member during Sprint 1 of the CryptoWise project. Work was divided based on each member's primary role as established in Iteration 0.

| Team Member | Role | Sprint 1 Tasks Completed |
|---|---|---|
| Rana Farhan Ijaz (23L – 0568) | Project Lead & Full-Stack Developer | • Next.js project setup and configuration<br>• Homepage section development<br>• Navigation bar with routing<br>• Overall frontend architecture<br>• API integration planning<br>• Deployment configuration<br>• Git repository management. |
| Muhammad Shaheer (23L – 0834) | ML Engineer & Data Scientist | • Binance WebSocket API integration<br>• Live price ticker implementation<br>• Real-time data streaming setup<br>• 24h price change calculation logic<br>• Color-coded indicator components<br>• Market prices page with real-time updates<br>• Search and filter functionality. |
| Muhammad Fahad Idrees (23L – 0533) | Backend Developer, Bot Specialist & Designer/Tester | • User authentication system<br>• JWT token implementation<br>• Password hashing with b crypt<br>• Responsive UI with Tailwind CSS<br>• Pages design and development<br>• Unit testing of authentication and UI components<br>• Sprint documentation. |

## 9.1 Sprint 1 Summary

Sprint 1 was completed successfully within the 2-week timeline. All 6 planned user stories were implemented and tested. The core platform is fully functional with user authentication, live price display, responsive UI, and all essential pages. The team is on track to begin Sprint 2 (ML Predictions & AI Assistant) as planned in the project roadmap.

| Sprint 1 Metric | Value |
|---|---|
| Total User Stories Planned | 6 |
| Total User Stories Completed | 6 (100%) |
| Total Estimated Hours | 52 hours |
| Stories Moved to Sprint 2 (Leftover) | None |
| Sprint Velocity | 52 story points |